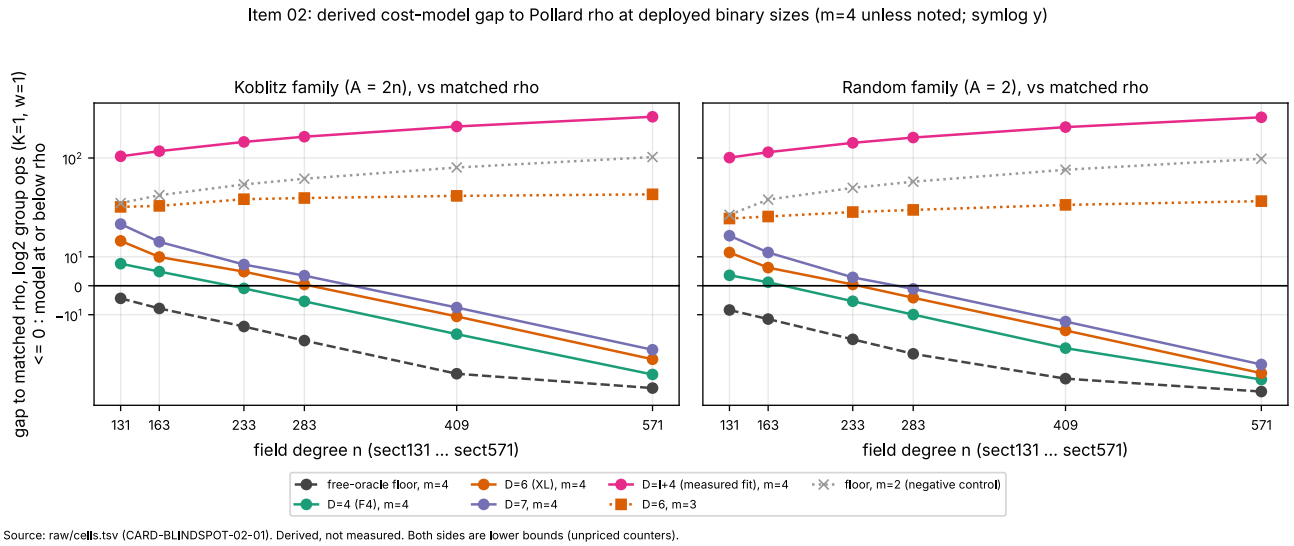
# Item 02: pricing binary curves at deployed sizes under the measured degree model

*Assessed 2026-10-18. All numbers are **derived** (closed-form model evaluation), none measured. Protocol: PROTOCOL.md (frozen). Cards CARD-BLINDSPOT-02-01 (sha256 96f35c67...) and CARD-BLINDSPOT-02-02 (sha256 f98dd5c6...).*

## Verdict

**ADVANCES (derived, model level), conditional on a constant-degree model; item-07 gate OPEN by the frozen rule; not a measured improvement.** Level: exponent (derived; arities m >= 4 have floors 2n/5 and n/3, below rho's n/2).

By the frozen rule some design arm flips (gap to matched rho <= 0) at K = 1: 100 design cells at K = 1, 31 of them at n = 571. Qualifications. (1) The flips come from the constant-D arms (D = 3, 4, 6, 7), which are hypotheses at deployed n; the measured degree evidence stops at n <= 15, l <= 5. The measured growth fit D = l + 4 never flips: gap +105.1 bits at n = 131 to +336.7 at n = 571 (Koblitz, w = 1, K = 1), a loss at every deployed n. (2) m = 3 never flips (non-floor gaps +14 to +72 bits), and the m = 2 negative control never crosses (minimum +24.50). (3) Both sides are lower bounds (unpriced counters), so no gap is a ranking, and a derived crossing is not an improvement over the index-calculus baseline.




Source: raw/cells.tsv (CARD-BLINDSPOT-02-01). Derived, not measured. Both sides are lower bounds (unpriced counters).

Figure 1: Gap to matched rho (log2 group operations, K = 1, w = 1) against field degree n, from raw/cells.tsv. Derived. Source: plot_gap.py. Run ids CARD-BLINDSPOT-02-01 / −02.

## Controls (all passed)

Positive: 30/30 and 16/16 committed rows identical (descent_degree_cost scale table; cost_d4 table); named rows 2^76.31 and 2^88.05; ECC2K-130 rho 60.8090. Range check: minimiser unchanged. Negative: m = 2 minimum gap +24.50 bits, never crosses. Monotonicity in w, K, D: 1392 checks, 0 violations.

## Gap to matched rho, w = 1, m = 4 (log2 group operations)

K = 1 is 1 GAE per column touch, K = 10 is 0.1, floor is 0 (AGENTS.md section 5a band). Negative: at or below rho.

| Koblitz | 131 | 163 | 233 | 283 | 409 | 571 |
|---|---|---|---|---|---|---|
| floor | −4.35 | −7.83 | −14.07 | −18.93 | −30.99 | −47.29 |
| D=4, K=1 | +7.62 | +4.89 | −0.92 | −5.38 | −16.70 | −31.67 |
| D=4, K=10 | +6.24 | +3.24 | −2.24 | −6.41 | −18.23 | −33.35 |
| D=6, K=1 | +15.50 | +9.94 | +4.87 | +0.44 | −10.58 | −25.38 |
| D=6, K=10 | +12.34 | +8.69 | +3.31 | −0.71 | −11.97 | −26.82 |
| D=7, K=1 | +21.30 | +15.17 | +7.33 | +3.49 | −7.52 | −22.10 |
| D=l+4, K=1 | +105.1 | +122.3 | +160.7 | +187.2 | +253.1 | +336.7 |

| Random | 131 | 163 | 233 | 283 | 409 | 571 |
|---|---|---|---|---|---|---|
| floor | −8.37 | −11.50 | −18.50 | −23.50 | −35.83 | −52.37 |

| | | | | | | |
|---|---|---|---|---|---|---|
| D=4, K=1 | +3.61 | +1.22 | −5.35 | −9.96 | −21.54 | −36.75 |
| D=4, K=10 | +2.23 | −0.43 | −6.67 | −10.98 | −23.07 | −38.42 |
| D=6, K=1 | +11.49 | +6.26 | +0.44 | −4.13 | −15.42 | −30.46 |
| D=6, K=10 | +8.32 | +5.01 | −1.12 | −5.28 | −16.81 | −31.90 |
| D=7, K=1 | +17.28 | +11.49 | +2.90 | −1.08 | −12.36 | −27.18 |
| D=l+4, K=1 | +101.1 | +118.7 | +156.2 | +182.6 | +248.3 | +331.7 |

Rho (log2 group operations), Koblitz: 60.81, 77.15, 111.39, 136.25, 198.99, 279.75; random: 64.83, 80.83, 115.83, 140.83, 203.83, 284.83. Sign unresolved inside the band: 5 of 480 design cells change sign between K = 1 and K = 10, 16 between K = 1 and K = 256; they are reported as unresolved, not as wins.

## Crossover n (scan, m = 4, w = 1, K = 1; above 571 is extrapolation)

| **model** | **Koblitz** | **random** |
|---|---|---|
| floor | 86 | 64 (grid edge) |
| D=3 | 191 | 143 |
| D=4 | 223 | 175 |
| D=6 | 292 | 238 |
| D=7 | 324 | 270 |
| D=6, w=2 | 533 | 475 |
| D=l+4 | none to 4096 | none to 4096 |

## Pricing and caveats

Unit: log2 group operations; oracle term = column touches / K. Unpriced on the attack side: Macaulay construction and memory traffic, root extraction, membership tests; on the rho side: orbit canonicalisation, partition hash, distinguished-point table. Both are lower bounds. Memory of the D=6, m=4, w=1 flips: 2^95 words at n = 409, 2^128 at n = 571 (rho O(1), not charged). The relay check (tools/task_card.py check-relay) reported both relays as not verbatim (missing lines listed in README); it is reported, not repaired; on-disk sha256 values equal the relayed digests and numbers were read from the files. Host L0 (shared container; irrelevant to a closed-form evaluation). Toy scale: degree evidence is measured to n <= 15, l <= 5 only. The model's free-oracle floor already crosses at n = 86, so it is optimistic for m >= 4. The crypto library does not compile at the branch base, so the instrument is std-only and built with rustc (deviation 8).